Low Voltage Grounded Auxiliary Power Unit (Draft)

AUTO1931 — Project Scope report

August 23, 2026

Overview

Formula SAE Electric (FSAE-E) vehicles have two main potential levels: one is the tractive system, which is up to 600 VDC, and the other is the low-voltage grounded (LVG) system, which is up to 60 VDC. RMIT Motorsports (RMIT-M) uses a 600 V tractive system and a 13.2 V LVG system. The tractive system is supplied by the tractive battery, which contains 143 lithium cells. Similarly, the LVG system is supplied by a lithium battery. However, it is a commercial unit certified by the FAA for aircraft!?

The low-voltage grounded auxiliary power unit (LVG-APU) is a proposed design to feed the LVG system from the tractive system while the tractive battery is installed in the vehicle. This APU would also have to be able to supply power while the tractive battery is disconnected from the vehicle, so it is also proposed that the LVG battery is scoped within the component. Hence the name "auxiliary power unit", as it provides the power required (LVG battery) during setup to close the contactor (similar to cranking amps), which is required to power the isolated (split grounds) DC-to-DC converter.

Later, the LVG battery buffers the line voltage during heavy loading, reducing the converter's required power output range. This also allows for the implementation of isolated external power input for the LVG battery, which is proposed to be an IP-rated (ingress protection) USB-C input to remove the dependency of another charging apparatus within the vehicle supporting equipment. This allows for a design with high self-reliance. At worst, it could be a USB-C laptop charger!

> **Supporting:** The impedance (L, R) between the LVG battery and the DC-to-DC converter is important due to the coupling (mutual influence) between them when supplying the line voltage across a wide range of power draws and rates of change of that draw. Hence the inclusion.

1 System Specification

The APU, as an intermediate component between the tractive system and the low-voltage grounded (LVG) system, is designed from the interfaces inwards (within reason). Hence the first specification is the input voltage range, which can be derived from the voltage cycle of a single lithium battery. The average rating is $\tilde{3}$.7 V, the maximum is $\tilde{4}$.2 V, and the minimum is $\tilde{3}$.0 V, with approximately 143 cells within the tractive battery. The input voltage range is $\tilde{6}$00–400 V, whereas the output, for simplicity, will be $\tilde{1}$2 V.

This leads into the power requirements, which is quite a complex question to answer in any capacity other than building and measuring the R27 LVG system. However, the R26 design can be used as a reference for this. According to an approximate current draw diagram, the LVG system should draw 15.9 A continuously with a peak of 66 A (Appendix 1a). This translates to a power output range of $\tilde{1}$90–790 W.

> **Supporting:** It should be noted that the exact technique used to approximate these values is unknown. And a few items are poorly defined.

The two topologies that were assessed were a flyback converter and an LLC converter. A flyback works by storing energy in the magnetic field of a transformer and then releasing it in pulses. This allows for the output voltage to be controlled at a high level by the duration of the repeating “on” pulse (PWM). The LLC converter uses a switching configuration to generate a square wave, and then that wave enters a resonant tank, which is simply a capacitor and inductor in series or parallel. This tank produces a sinusoidal wave and, like the flyback, has a transformer which then outputs the energy (at a high level). The output can then be controlled through the switching signal by varying frequency and timing.

From those two topologies, the main cornerstone for selecting one is the power output required rather than specifically implementation complexity. From this perspective, the LLC is preferable as it often has better scaling capabilities. Since it is not storing energy in pulses, the maximum output range can be much higher. The LLC also has two different methods of controlling the output, each of which has different power ranges. This allows for more implementation options on top (half-bridge vs. full-bridge). Hence, the LLC topology was chosen.

Therefore, the LLC converter will need to be sized for the approximate nominal power draw and acceptable current density within the LLC transformer. Using a rule-of-thumb model (Appendix 1b) based on current density and wire diameter, a series of graphs relating current density and wire diameter were generated. From that and the power range inherited by the R26 documentation and R&D report about future driverless integration (Appendix 1c), a final requirement of 300 W was decided. It should allow up to 33% output buffer based on R26 continuous draw, while also not making the transformer impractical to design.

> **Rational:** At a current density (current per cross-sectional area) of $\sim 2.6\,A/mm^2$ on the secondary with a ratio from an RMIT-M paper of 21:1, a bulk (solid) diameter of 3.5 mm is required, which is reasonable. It falls within the natural cooling range at 2.0–3.0 A/mm^2 and insulation class A, which is abundant.

2 Literature and Design Review

The search performed during this scoping phase did not identify the LVG-APU (APU) as an established discrete system architecture. Therefore, the review focuses on the isolated LLC converter (LLC), which forms the centre of gravity of the proposed architecture.

```
Traction battery (600 V)
        ↓
LVG-APU Interface (Undecided Connector)
-----------------------------------------------------------------
Isolated LLC resonant DC/DC (300 W Target) ← (We're Here)
        ↓
LV battery buffer (Energy Buffer) ← Isolated Charger (External Input)
-----------------------------------------------------------------
LVG-APU Interface (Undecided Connector)
        ↓
LV vehicle systems (12 V) (200 W Estimate)
```

Important: We're here! Given the initial scoping of "600V to 12V Isolated DC/DC Converter & LV Integration," the above analysis had to be done to decompose and reconstruct the system for clear design and objectives.

2.1 A Customised Low-voltage Power Supply for a Formula SAE Electric Racing Vehicle

This paper is relevant as it describes the design and implementation of a 240 W LLC converter for FSAE and was completed in 2024 by RMIT-m students for a capstone. This design implements the LLC converter as a single package consisting of five different boards following a modular approach. The switching method is a half-bridge and gate driver, which is then controlled by a resonant controller. The switches were 1200 V silicon carbide FETs, and the frequency range was 100–200 kHz with a 21:1 turn ratio within the transformer and series capacitor. The output is then filtered by a synchronous rectifier.

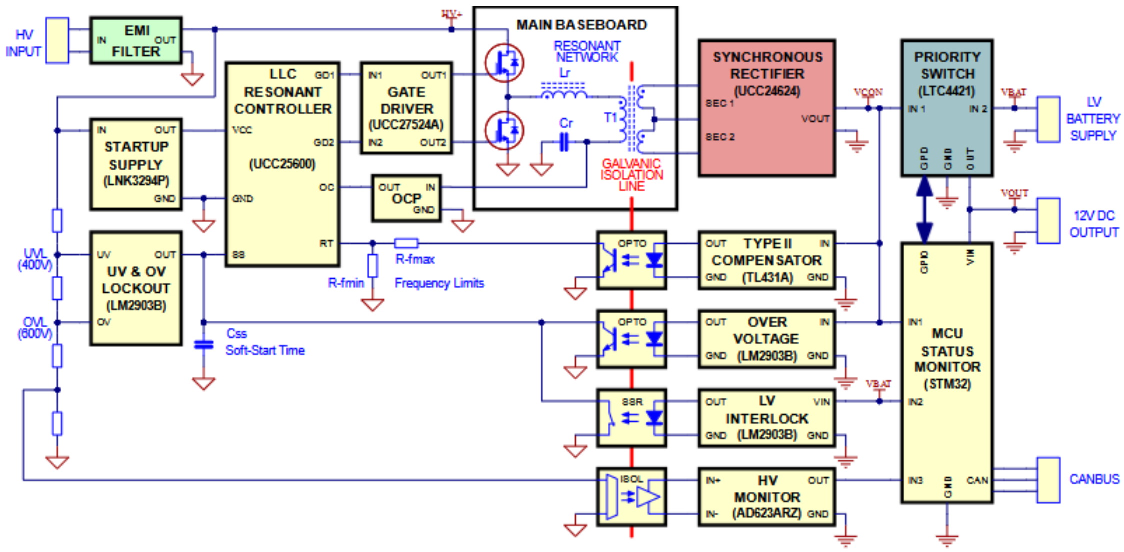

Figure 1: Block diagram of the LLC converter from the paper

2.2 DCDC: An open source credit card sized DC/DC converter to convert up to 600 V to 24 V in an energy-, cost- and space-efficient way.

This repository (repo) contains a high-voltage DC/DC converter designed for the FSAE German competition and, according to the repo, passed scrutineering twice and has been operating without issues for over 6 months. The maximum output power is 750 W for 60 seconds and 500 W continuous. The design input voltage range is 200 to 600 V and has an output range of 24 V, with the ability to customize it for 12 V to 48 V outputs. The design is quite small at only 167 grams and has a volume of 0.208 L.

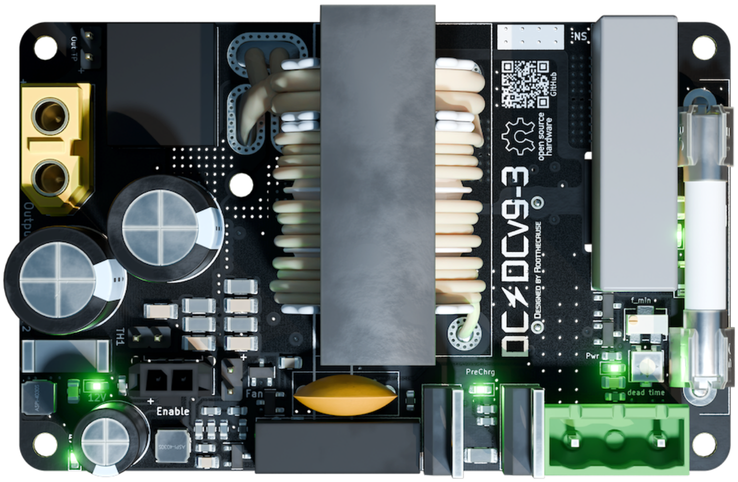

Figure 2: DCDC PCB design populated with components.

According to the repo, the design features a solid-state precharge circuit, active rectification, and under- and over-voltage protection. It also has over-current protection and, lastly, HV and LV fuses. The repo notes that the dynamic operating range is 90 to 200 kHz to 'ensure high efficiency for various loads' and 'good EMI compliance due to . . . LLC the topology'

2.3 Interpretation

The two reviewed designs demonstrate that isolated LLC conversion is an established approach for high-voltage to low-voltage conversion. Both operate over an input range overlapping with the 400–600 V requirement, while their power ratings of 240 W and 600 W provide precedent for the 300 W target.

However, neither design directly establishes the proposed APU architecture. The reviewed systems seem to treat the DC/DC converter as the power-supply component in abstract rather than as a coupled system. Therefore, these designs will be used as precedent for the LLC subsystem rather than as a complete implementation reference.

The 2024 RMIT-M paper will be the primary reference for the LLC converter design and implementation, as it shows the design methodology for designing the system using first harmonic approximation (FHA) and detailed transformer construction methods. The modular design approach is quite elegant from a testing perspective. Hence a similar approach will be adopted for this implementation.

3 Conceptual Design

The proposed design for the LVG-APU (APU) is to began with the LLC converter as two independent boards, which are then mechanically and electrically connected via the 'high-frequency transformer'. This reflects the natural system boundaries between the high and low voltage sides while also improving the ability to perform sub-circuit testing following a similar approach to the 2024 RMIT-M paper.

High Level Topology:

```
Interface (Undecided Connector)
LLV-HVS - Tractive battery input (400-600V domain) (Unknown EMI)
-----------------------------------------------------------------
EMI factor (Common-mode chokes) (Shunt Capacitors)
         ↓
Half Bridge Mosfets ← Half bridge Driver IC ← LLC Controller ← Optocoupler
         ↓
Resonant Tank Circuit
-----------------------------------------------------------------
High Frequency Transformer (Primary) (600 AC) (Unknown Ripple)

=============================================
Ferrite Core (Magnetic & Structural Coupling)
=============================================

LLV-LVS - High Frequency transformer (Secondary) (12V AC) (Unknown Ripple)
-----------------------------------------------------------------
Synchronous Rectifier  → Status MCU (STM32) ← Optocoupler
         ↓
Integrated battery (12 V) (Undecided Capacity) ← Isolated Charger
-----------------------------------------------------------------
LV-Interface (Undecided Connector) (12V Domain) (Unknown EMI)
```

This design follows the 2027 Formula SAE rules draft. It allows for a grounded low-voltage (GLV) system to operate independently of the tractive battery (EV.4.4.1). It also is galvanically isolated via magnetic coupling and optocouplers for signal routing (EV.7.5.1). It also avoids the same-board tractive system and GLV rule (EV.7.5.7) with two independent boards; however, for the optocouplers on the LLC-HV board, the area will be clearly marked and have a clearance distance of 3.0 mm and a creepage distance under conformal coating of 4.0 mm when operating at 300–600 VDC.

The implementation plan is firstly to use planar finite element analysis to design the ferrite core transformer, and then design the LLC-HV board. The transformer will then be characterized, and an appropriate capacitor will be chosen. Then the LLC-HV board and transformer will be tested together with a dummy load. After that, the LLC-LV board will be made based on those testing results. The LLC-HV and LLC-LV boards will then be tested together and characterized, with the transient response depending in part on the capability of the APU-integrated battery. After which, the whole system will be packaged together as the LVG-APU and tested for EMI compliance, etc.

4 Project Management

The project will follow the waterfall structure with each deliverable within the APU providing information for the next. The project deliverables consist of the LLC-HVS and LLC-LVS board. The transformer, LVG battery and finally the APU packaging around the whole system. Please reference the gantt Chart below to under the design cycle of the LVG-APU.

As like the work done on for scoping this project, all engineering notes and related information can be found on the projects Github page.

5 Conclusion

The LVG-APU for the R27 vehicle will be a first iteration of a new power system within the vehicle. The design arises from the architectural requirements of the LVG system. And the requirements formed arise from current implementation and reference designs described in the literature review area. The system consist of 5 deliverables: the LLC-HVS and LLC-LVS boards, the transformer, LVG battery and the APU packaging.

6 References

7 Appendices